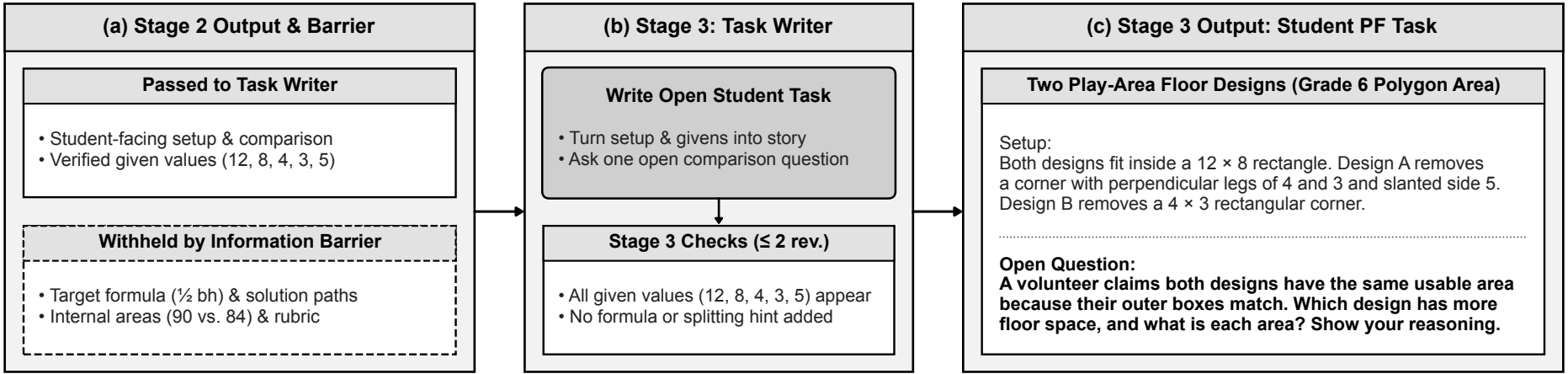
(a) Stage 2 Output & Barrier
Passed to Task Writer
• Student-facing setup & comparison
• Verified given values (12, 8, 4, 3, 5)
Withheld by Information Barrier
• Target formula (½ bh) & solution paths
• Internal areas (90 vs. 84) & rubric
(b) Stage 3: Task Writer
Write Open Student Task
• Turn setup & givens into story
• Ask one open comparison question
Stage 3 Checks (≤ 2 rev.)
• All given values (12, 8, 4, 3, 5) appear
• No formula or splitting hint added
(c) Stage 3 Output: Student PF Task
Two Play-Area Floor Designs (Grade 6 Polygon Area)
Setup:
Both designs fit inside a 12 × 8 rectangle. Design A removes a corner with perpendicular legs of 4 and 3 and slanted side 5. Design B removes a 4 × 3 rectangular corner.
Open Question:
A volunteer claims both designs have the same usable area because their outer boxes match. Which design has more floor space, and what is each area? Show your reasoning.